C18 Rate Equations – Knowledge Organiser

AQA A-level Chemistry • Set 1: Rate Equations and Orders **Paper 2**

#	Question	Answer
1	General rate equation	rate = $k[A]^m[B]^n$
2	Order with respect to a reactant	Power of its concentration in the rate equation
3	Overall order	Sum of the individual orders
4	Only factor that changes *k*	Temperature
5	Effect on rate of changing a zero-order concentration	None
6	Order if doubling concentration doubles rate	First
7	Order if doubling concentration quadruples rate	Second
8	Rate-determining step	Slowest step in the mechanism
9	Species that appear in the rate equation	Those in or before the rate-determining step
10	Units of rate	mol dm^{-3} s^{-1}

C18 Rate Equations – Knowledge Organiser

AQA A-level Chemistry • Set 2: Units of k and Arrhenius **Paper 2**

#	Question	Answer
1	Units of *k*: zero order	$mol\ dm^{-3}\ s^{-1}$
2	Units of *k*: first order	s^{-1}
3	Units of *k*: second order	$mol^{-1}\ dm^{3}\ s^{-1}$
4	Units of *k*: third order	$mol^{-2}\ dm^{6}\ s^{-1}$
5	Arrhenius equation	$k = Ae^{-Ea/RT}$
6	Name of *A* in the Arrhenius equation	Pre-exponential (frequency) factor
7	Log form of the Arrhenius equation	$\ln k = -E_a/RT + \ln A$
8	Gradient of ln *k* against 1/*T*	$-E_a/R$
9	y-intercept of ln *k* against 1/*T*	$\ln A$
10	Units of *T* in the Arrhenius equation	K

C18 Rate Equations – Knowledge Organiser

AQA A-level Chemistry • Set 3: Graphs **Paper 2**

#	Question	Answer
1	Concentration–time graph: zero order	Straight line, negative gradient
2	Concentration–time graph: first order	Curve with constant half-life
3	Rate–concentration graph: zero order	Horizontal line
4	Rate–concentration graph: first order	Straight line through origin
5	Rate–concentration graph: second order	Upward curve through origin
6	Half-life	Time for concentration to halve
7	Half-life of a first-order reaction	Constant
8	Initial rate from a concentration–time graph	Gradient of tangent at $t = 0$
9	Initial rate in a clock reaction is proportional to	$1/t$
10	Value of R	$8.31\ \text{J K}^{-1}\ \text{mol}^{-1}$